

CHAPTER 7

Indirect Functional Behavioral Assessment

Once you eliminate the impossible, whatever remains,
no matter how improbable, must be the truth.

—SHERLOCK HOLMES (by Sir Arthur Conan Doyle)

Conducting an FBA is a complex task that involves consideration of a wide range of variables. Before initiating an FBA, it is important to recognize that it is both a sequential and simultaneous process. It is a sequential process because we tend to assess behavior by marching through a logical series of hypothesis-testing investigative procedures. It is a simultaneous process because we are continuously considering a host of variables that may be contributing to the occurrence of interfering behaviors.

Consider the visual image depicted in Figure 7.1. Here, as the school practitioner begins the FBA process, she is trying to remember each of the variables and practices that must be considered when conducting a comprehensive FBA. She is finding this to be a daunting task that requires a tremendous amount of focus and organization. We've been there. In fact, in the old days, we often initiated an assessment armed with only a clipboard, graph paper, a pencil, and our own ingenuity. We invariably neglected to ask specific questions and overlooked potential contributing variables. You've probably done that too. For example, 30 minutes after the initial interview with the classroom teacher, while driving to a student assistance team (SAT) meeting at another school, you think, "Oh no! I forgot to ask about Jerry's skill deficits!" and "It sounds like Jerry's disruptive behaviors are motivated by teacher attention, but wait . . . I forgot to ask how other students in the class respond when he engages in disruptive behavior." Then, 45 seconds later, after you regain your composure from nearly running a red light because you were not concentrating on your driving, you reflect, "Holy cow! What about the difficulty of the tasks? Perhaps disruptive behavior is motivated by escape/avoidance of specific tasks due to an underlying specific learning disability?"

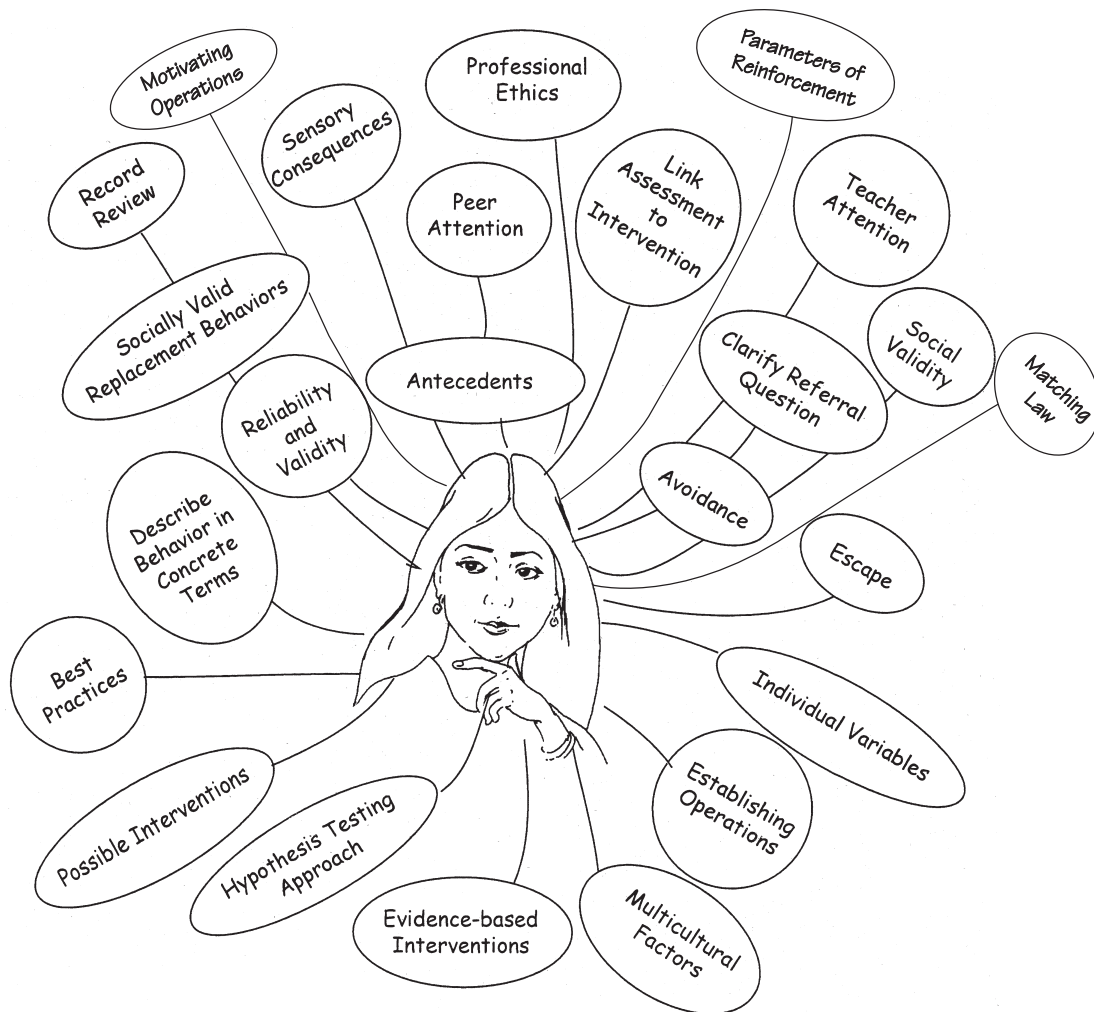


FIGURE 7.1. All things considered . . . in the professional life of a school-based evaluator.

In our experience, FBAs are more efficient and accurate when evaluators adopt a problem-solving process and adhere to a game plan that includes (1) a conceptual framework rooted in the principles of applied behavior analysis and (2) the necessary tools to conduct the assessment. In Chapter 6, we provided you with the necessary conceptual framework: the BAPS model. Throughout the FBA process, from the initial record reviews and interviews to the final treatment analyses, it is important to continuously assess and reassess interfering behaviors against the backdrop of the BAPS model. You need to keep an open mind, consider the full range of variables that influence behavior, and engage in hypothesis testing to “rule in” and “rule out” the relevant contributing factors. In Chapters 7–9, we review a variety of assessment procedures to address the second key aspect of the game plan: having adequate resources, or tools, to conduct the FBA. Every student and situation is different; therefore evaluators need a wide variety of tools at their disposal to get the job done efficiently and effectively. FBA is not a one-size-fits-all process. You need to carefully consider the referral question and all available tools or assessment procedures before developing a game plan that will lead to a valid FBA. A comprehensive FBA does not

require application of all available assessment tools; however, a comprehensive FBA always includes some level of indirect behavioral assessment. In the remainder of this chapter, we describe and illustrate several indirect FBA procedures.

INDIRECT FBA

Indirect assessment is so named because information regarding antecedents, consequences, and other critical variables is gathered *indirectly* via record reviews, rating scales, interviews, and the like. The data gleaned from these measures are generally not as reliable as those obtained from more direct procedures and hence are considered to offer useful *adjunct* information. Indirect assessments are typically the first procedures used in an FBA because they help evaluators begin developing hypotheses about the function(s) of interfering behavior and inform the design of meaningful descriptive and/or experimental assessments. We have found that the indirect FBA procedures described in this chapter are useful for (1) identifying and describing interfering behaviors; (2) identifying and describing antecedent, individual, and consequence variables that may influence the occurrence of interfering behaviors; and (3) generating testable hypotheses regarding the function(s) of interfering behaviors. Essentially, the indirect FBA is the first stage of the assessment process, and the resulting data guide subsequent hypothesis-testing methods.

Considerations for Evaluators Conducting Indirect FBAs

The entire FBA process is a hypothesis-testing approach. It is an investigative process that involves gathering data, identifying potential relationships between environmental variables and interfering behavior, collecting more data, analyzing the influences of environmental variables on interfering behavior, and confirming or disconfirming hypotheses. Remember, hypotheses are not etched in stone. Many times they are more like drawings in the sand—with the next wave of data washing them away. We must always keep in mind that indirect FBA procedures gather information from indirect sources, and we need to exercise critical thinking skills when interpreting these data. Because records, rating scales, and interviews rely on the reports of others, it is possible that practitioners may arrive at inaccurate conclusions unless they *approach interpretation with healthy skepticism*. We believe that it is crucial to consider all existing data, rating scale results, and student/teacher/parent perspectives, while practicing *philosophical doubt*, as you formulate your hypotheses.

INDIRECT FBA PROCEDURES

During the indirect phase of an FBA, we recommend using the following procedures:

- Review of records
- Analysis of rating scale results
- Behavior-Analytic Problem-Solving Interview (BAPS-I)
- Behavioral Stream Interview (BSI)

In the subsequent sections of this chapter, we review each of these indirect assessment procedures, discuss considerations for their use, and offer tools and resources to support application.

Review of Records

To improve efficiency, it is extremely important to conduct a thorough record review in the early stages of the FBA process. Cumulative educational records contain valuable background information about students' personal characteristics, skill strengths and weaknesses, behavioral presentation, and history of services. Table 7.1 summarizes the type of information we look for during record reviews and offers a rationale for considering each type of information within the context of an FBA.

Analysis of Rating Scale Results

Although traditional psychological rating scales provide no indication of the function(s) of interfering behaviors, information from these tests may suggest underlying personal characteristics and/or skill deficits that contribute to the occurrence of interfering behaviors. As part of the diagnostic process, school psychologists often ask students, teachers, and caregivers to complete behavior rating scales to determine whether a student exhibits clinical levels of symptoms associated with particular syndromes or disorders. School psychologists also may administer adaptive, social skill, or executive skill rating scales to gather information about students' unique strengths and weaknesses. As we discussed in Chapter 6, information about personal traits and adaptive, social, or emotional skill deficits can facilitate an understanding of why some situations are particularly challenging and likely to evoke interfering behavior. Rating scales serve as a valuable source of information about these associated individual variables. Take, for example, a student whose rating scale data indicate that he has clinical levels of anxiety, specifically social anxiety. In the context of an FBA of class refusal behavior, an evaluator may hypothesize that the autonomic arousal and worries about social judgment typically associated with social anxiety may serve as motivational triggers that make avoiding classes with social interaction more likely.

Practitioners conducting FBAs also may use a special type of rating scale, one whose sole aim is to assess the function(s) of interfering behaviors. Unlike typical psychological symptom-based rating scales, function-focused rating scales ask informants to rate the extent to which an identified interfering behavior occurs under certain conditions. For example, an item on a function-focused rating scale may ask whether the interfering behavior occurs when a difficult task is presented. If an informant indicates "yes" to this and similar items, it suggests that the function of the behavior may be escape/avoidance. Several function-focused rating scales have been developed since the 1980s (Cooper et al., 2007). Widely used scales include the Functional Analysis Screening Tool (FAST; Iwata & DeLeon, 1996); the Questions About Behavioral Function checklist (QABF; Matson & Vollmer, 1995); and the Motivational Assessment Scale (MAS; Durand & Crimmins, 1988). Although these tools are efficient and convenient, it is important to remember that the reliability and validity of the data may be questionable (see the box below). Therefore, we recommend considering the use of these tools only as preliminary assessments to develop hypotheses for direct evaluation.

TABLE 7.1. Record Review Guidelines

Information category	What to look for?	Relevance to FBA?
Social history	Social stressors such as changes in address and school placements, foster home or residential placements, parental separations and marriages, or death of family members or friends.	Social stressors may point to possible MOs for interfering behavior.
Medical history	Developmental delays; vision, hearing, or motor impairments; current medications; and other medical issues (e.g., long-term illness, head trauma, enuresis)	Medical conditions may suggest relevant personal characteristics, skill delays/deficits, and possible MOs for interfering behavior.
Attendance history	Patterns and total number of absences	Frequent absences may indicate the presence of skill deficits associated with lack of instruction and patterns of absences may provide clues about possible MOs for interfering behavior.
Disciplinary history	Topographies of interfering behavior, times and locations associated with interfering behaviors, disciplinary actions imposed, and trends in the frequency or intensity of interfering behavior	Disciplinary records may chart the progression of interfering behavior, provide clues about potential antecedents and consequences for interfering behavior, and offer information about effective and ineffective disciplinary strategies.
Academic history	Historical and current scores on state-mandated standardized assessments, standard- or competency-based grades	Test scores and performance data highlight academic skill delays/deficits and which content areas or academic tasks may be particularly challenging (possible MOs for interfering behavior).
Previous evaluation reports	Data from prior FBA, psychoeducational, academic, speech and language, occupational therapy, and other evaluations	Evaluation reports provide information about relevant personal characteristics and skill delays/deficits, and FBA reports reveal historical antecedents and consequences for interfering behavior and potential changes in function over time.
Previous intervention reports	Progress monitoring data from academic, social–emotional, or behavioral interventions	Progress monitoring data identify interventions that have been successful and unsuccessful. (If successful, why are they not currently being used? If unsuccessful, why are they continuing to be used?)
IEP	Services, supports and accommodations; instructional goals and objectives, instructional methods and monitoring systems, progress data, etc.	IEP documents reveal skill deficits and the degree to which interfering behaviors are being addressed and monitored within the school setting.

Not So FAST!

A word of caution about the FAST and other functional assessment rating scales. Evaluators may naively believe that if they hand out enough of these rating scales, they can confidently determine the function of an interfering behavior. While these rating scales are a very efficient method of identifying behavioral functions, there is no such thing as the 4-minute FBA.

Iwata, DeLeon, and Roscoe (2013) examined the validity of the FAST and found that it accurately classified the function of interfering behavior only 63% of the time. At his ubiquitous functional analysis trainings, Iwata dissuades practitioners from relying on the FAST. His recommendation is to move to direct functional assessment of behaviors as FAST as possible.

Interviews

The most important part of the indirect FBA process involves interviewing individuals who have observed the interfering behavior. Through systematic interviews, practitioners can gain valuable descriptions of interfering behaviors and associated contexts, personal characteristics, skill delays/deficits, antecedents, and consequences. In turn, descriptions of these variables allow practitioners to develop hypotheses about behavioral function(s) for subsequent verification.

Potential Errors

Before we launch into a description of interview procedures, we think it is important to review the potential pitfalls you may encounter. There is no doubt that a skillfully conducted interview can shed light on behavioral function; however, this information must be

Fake Function

As a member of the SAT, a school psychologist was asked to conduct an FBA and develop a behavior support plan for Jerry, an adolescent with ASD who engaged in aggressive behaviors (e.g., hair pulling, hitting, and biting others). When the school psychologist asked school staff to offer an example of the aggressive behavior, a paraprofessional described an incident that began when she asked Jerry to fold some towels. The paraprofessional explained that she introduced the expectation by saying, "Jerry, it is time to fold towels. These are towels you have already washed and dried. I am sure that you can fold these towels. You have folded towels before. Do you want help, or can you do these all by yourself?" She reported that Jerry then immediately jumped out of his chair, lunged toward her, grabbed her hair, and wrestled her to the floor. This resulted in a physical restraint that lasted 44 minutes. The paraprofessional stated that the reason Jerry engaged in aggressive behavior was his clear and obvious dislike of folding towels. Some practitioners may conclude at this point that they understand the function of the behavior because (1) they have a firsthand description of the sequence of events that led to the physical restraint, (2) it is a reasonable hypothesis that Jerry reacted in such an aggressive manner because he obviously dislikes folding towels, and/or (3) they are being pressured to design a behavior plan quickly due to the dangerous nature of the aggressive behavior. The school psychologist made this very conclusion.

Although, on the surface it seemed that the expectation to fold towels caused Jerry to become aggressive, the school psychologist made an *error of association*. After several intervention attempts focused on reducing the difficulty and nature of the towel-folding task, Jerry continued to display aggression. Further assessment revealed that it was not towel folding but vocal cues that triggered Jerry's aggression. The school psychologist in this situation learned from experience that just because something like towel folding is *associated* with a behavior, it does not mean it is a *cause* of the behavior.

carefully scrutinized during the FBA process to avoid faulty conclusions. There are many errors that may arise from interview data. Consider the scenario in the box on page 111.

The *error of association* committed by the school psychologist in the previous example is only one type of error that may occur during an indirect FBA interview. Other types of errors include:

- Recency error of perception
- Primacy error of perception
- Error of exaggeration
- Error of generalization

The *recency error of perception* occurs when interviewees attribute the cause of an interfering behavior to the variables present during the most recent occurrence. This error is illustrated by the teacher in the following example:

REGENCY ERROR EXAMPLE

Phil is a fourth-grade student in a regular education class. He was referred for an FBA because of frequent disruptive outbursts that sometimes included swearing and minor property damage. During an FBA interview, the classroom teacher indicated that Phil's outbursts were the result of an abrupt transition from a relatively unstructured, highly physical activity (e.g., recess or gym class) to the classroom. She reported that the most recent behavioral incident had occurred within 15 minutes after gym class and that most of his outbursts had occurred after gym or recess. To be certain, the school psychologist conducted additional assessments to examine the teacher's hypothesis and identify other potential triggers for Phil's outbursts. The school psychologist conducted several direct observations of Phil in his classroom, both immediately after recess and gym and at other randomly selected times. The data revealed two findings that were significant for understanding Phil's outbursts: (1) his outbursts occurred after recess or gym only occasionally and (2) more than 65% of his outbursts occurred immediately after his teacher provided critical feedback about the quality or the quantity of his academic work.

The *primacy error of perception* occurs when the cause of an interfering behavior is attributed to variables that were present the very first time the interviewee observed it. Like the recency error of perception, the primary error of perception may lead to misleading conclusions about the function(s) of interfering behaviors. The following example illustrates this potential error:

PRIMACY ERROR EXAMPLE

Sheryl is an eighth-grade student receiving special education services. She was diagnosed with specific learning disabilities in math and reading, and presents with significant deficits in social skills. She often threatens her classmates with physical harm, although she has not yet reached the point of physical aggression. When a member of the SAT interviewed one of Sheryl's teachers, the teacher expressed certainty about the reason for her verbal

aggression: teasing from classmates regarding her academic performance. To support her hypothesis, the teacher recalled the first time she directly witnessed Sheryl threatening a classmate, which was about 10 weeks prior to the interview. The student had teased Sheryl about her inability to read a selected paragraph, and Sheryl responded by threatening to “beat her up” if she did not stop teasing her. The teacher concluded by saying that, although she did not always hear Sheryl’s interactions with peers, she was convinced that teasing was the reason for Sheryl’s verbally aggressive behavior. In this example, the teacher’s description of an incident 10 weeks prior may have been accurate, but it may or may not have been representative of the events that were continuing to trigger verbal aggression. Sometimes the apparent triggers for the initial occurrence of an interfering behavior (teasing, in this case) stand out so much in memory that informants assume similar events continue to function as triggers. However, without conducting a comprehensive FBA, it is impossible to draw firm conclusions about the function(s) of behavior.

The *error of exaggeration* occurs when interviewees overstate the frequency, duration, and/or intensity of behavior. Red flags for this error include words and phrases such as *always*, *never*, *constantly*, *the worst ever*, and *completely out of control*. Consider the following example:

ERROR OF EXAGGERATION EXAMPLE

Floyd is a student who receives special education services under the category of emotional disturbance (ED). He was referred for an evaluation because he displays oppositional–defiant behaviors (e.g., verbally refusing to complete academic assignments, throwing books, and ripping up assignments) in both general education and resource classrooms. During an FBA interview, one of Floyd’s teachers stated, “He *always* misbehaves when I ask him to do his work.” When asked to elaborate, the teacher said, “*Every time* I ask him to complete an assignment, he throws one of his tantrums.” The school psychologist conducted three observations within each of the classroom settings and found that there was not a perfect correspondence between directions to complete assignments and interfering behaviors. In fact, further investigation revealed that the conditional probability of oppositional–defiant behavior given an instruction to complete an assignment was only 30%. Although there may be times when a behavior always follows an antecedent, it is our experience that words and phrases such as *always* and *every time* signal the need for further investigation.

The *error of generalization* occurs when interviewees or evaluators assume the same functional hypotheses apply across students or behaviors. The *error of generalization across students* is evidenced when team members assert that the variables contributing to the interfering behaviors exhibited by one student are the same variables contributing to similar interfering behaviors exhibited by another student. For example:

ERROR OF GENERALIZATION ACROSS STUDENTS EXAMPLE

A recent FBA suggested that Jimmy engages in swearing behavior to gain attention from peers. When Carter begins swearing, the school team assumes that his behaviors also func-

tion to gain access to peer attention. In this case, the similarity of interfering behaviors across students led team members to believe that the function of swearing was the same for both students. However, it is entirely plausible for swearing to be reinforced by peer attention for one student and by escape from difficult assignments for another student. Without conducting a separate FBA of swearing with Carter, it is not possible to draw conclusions about the functions of his behavior.

The *error of generalization across behaviors* is evidenced when team members assume that the functional antecedents and consequences for one interfering behavior influence the occurrence of all interfering behaviors. Consider the case of Erin:

ERROR OF GENERALIZATION ACROSS BEHAVIORS EXAMPLE

Erin is a student who displays multiple interfering behaviors, including aggression, property destruction, swearing, and hand biting. Indirect FBA (interview) results suggested that Erin's interfering behaviors are reinforced by escape from difficult tasks. Subsequent function-based interventions were successful in treating aggression and property destruction, but levels of swearing and hand biting behaviors increased over time. The evaluator therefore revisits the assessment process and conducts a descriptive FBA. The resulting data suggest that swearing and hand biting are maintained by social attention from teachers, not escape. In this case, FBA interviews led to an error of generalization; the interviewees assumed that all topographies of Erin's behavior were related to the same environmental variables, and the evaluator did not gather data using multiple methods to ensure convergence of evidence.

Interview Guidelines

The most effective FBA interviews are not formulaic. Using a standard interview form with the exact same set of structured interview questions for every assessment is not recommended. Instead, it is recommended that evaluators employ flexible interview procedures that fit the needs of each unique assessment situation.

When conducting FBAs, we recommend using a combination of semistructured and unstructured interview procedures. For example, an initial interview may be conducted with a teacher or paraprofessional using a comprehensive semistructured interview tool such as the BAPS-I (Form 7.1) to guide the conversation. Semistructured interview tools like the BAPS-I include a comprehensive range of predetermined questions and recommended prompts, but questions may be added or omitted as appropriate. Subsequent interviews with the same informant, others familiar with the referred student, and/or happenstance witnesses may be relatively unstructured to gather detailed information about specific behavioral incidents. The BSI, for example, takes the form of a stream of unstructured questions and answers. The BSI focuses on identifying and describing four central components of the BAPS model—context, MOs, S^Ps, and reinforcing consequences—associated with a specific occurrence of interfering behavior. Information from the BSI may be captured using a narrative “flow chart” and then mapped onto the Incident-Based Functional Assessment Form (IBFAF; see Figures 7.2 and 7.3 for examples).

Behavior-Analytic Problem-Solving Interview

The BAPS-I (Form 7.1) should be conducted with individuals most familiar with the student and the circumstances of his or her interfering behavior. The purpose of this semistructured interview is to identify and describe the variables that contribute to occurrences of interfering behavior. The interview process results in the identification and description of an interfering behavior as well as *hypotheses* pertaining to behavioral function. For FBAs addressing multiple interfering behaviors, we recommend a separate interview (and forms) for each behavior.

The BAPS-I begins with questions about the student's unique strengths. Opening the conversation in this manner facilitates the establishment of rapport with the interviewee. We think of it as a "warm-up" strategy to establish momentum with information sharing. Initial discussions about the student's strengths also provide valuable information about the student's preferences and current repertoire of skills—both of which are essential considerations when developing teaching and reinforcement-based strategies for inclusion in a comprehensive behavior intervention plan.

The next portion of the BAPS-I cues the interviewer to identify and define the interfering behavior. Although parents, teachers, and other school staff are usually quite good at *identifying* behaviors that are problematic, they often experience considerable difficulty *describing* interfering behavior in concrete terms. For example, a teacher may describe a student's behavior as "noncompliant," which is a label that means different things to different people. The job of the interviewer is to engage with the interviewee to construct a definition for "noncompliant" that can be immediately recognized by anyone who observes the behavior. When interviewees have difficulty describing an interfering behavior, we sometimes ask them to "act out" or model it. This provides us with a visual image of the interfering behavior, which assists with the development of a written description. A word of caution is needed here. We want to emphasize that we do not *always* ask interviewees to act out the behavior in question. There are times when you most certainly would *not* want to request a demonstration. Why, you ask? Well, some behaviors may be dangerous to self or others (e.g., aggression, property damage, ruminative vomiting) or inappropriate to model (e.g., sexual behaviors). The key here is to arrive at a clear understanding of the target interfering behavior to ensure that the interviewer and interviewee are speaking a common language. The subsequent sections of the BAPS-I focus on identifying individual and environmental variables that seem to influence the target interfering behavior. If the evaluator proceeds

Defining Interfering Behaviors

Before proceeding with descriptive FBA procedures, it also is essential to define interfering behaviors in such a way that two observers would agree on the occurrence or nonoccurrence of the behavior after reviewing a written definition. As we discuss in Chapter 8, it is usually helpful to conduct an anecdotal observation of the referred student before formalizing the operational definition for the interfering behavior and designing matched data collection procedures. For example, after conducting an observation an evaluator may refine the interview-generated description of "hitting peers" into the operational definition of "striking a peer with an open palm from a distance of at least 3 inches." This process of refining behavioral definitions is a normal part of the sequential and simultaneous process mentioned at the beginning of the chapter.

with interview questions before solidifying a specific and objective description of the target behavior, then interviewees may offer responses that are irrelevant (e.g., describing antecedents and consequences associated with a different interfering behavior), which leads to inaccurate hypotheses.

The remainder of the BAPS-I is formatted to help interviewers collect critical (*relevant*) information about the individual and environmental variables *directly associated with the target interfering behavior*. This point is critical. Referred students often present with multiple skill deficits and experience behavioral challenges in multiple situations. The goal of the FBA interview is to hone in on the variables *most relevant* to understanding the target interfering behavior. Accordingly, interviewers must be prepared to (1) frame questions in reference to the target interfering behavior (e.g., “Given an occurrence of physical aggression directed toward a peer, how do school staff respond?” or “Given a writing assignment that typically triggers noncompliance, what skill deficits may be contributing?”) and (2) address “drift” by redirecting interviewees to focus on one target interfering behavior at a time.

The BAPS-I form includes open-ended questions designed to elicit information from interviewees about each of the variables included in the BAPS model. These questions facilitate the collection of comprehensive information in a logical sequence. Depending on how interviewees respond, though, interviewers may need to jump around the form to record information. You will notice that the form is organized into separate sections pertaining to the interfering behavior, the context, MOs, S^Ds, reinforcing consequences and parameters of reinforcement, personal characteristics, and relevant skill delays/deficits. Each section includes recommended questions in **bold** font to stimulate initial discussion. Below these starter questions are supplemental prompts (questions and/or considerations printed in *italics*) for interviewers. These supplemental prompts are included to help interviewers probe for additional information, as needed, to gather comprehensive data about each variable. Finally, each section includes a table for recording information provided by interviewees. When completing the interview and recording results, it is important to keep in mind a few key points:

1. *It is not necessary for interviewers to ask every question.* For example, if early responses to interview questions indicate that the target interfering behavior occurs exclusively in the context of recess and unstructured social activities, then it is not necessary (or logical!) for the interviewer to probe for information about the types of academic tasks that are challenging for the student.

2. *Prompts are included to help interviewers formulate relevant individualized questions.* Some prompts include optional questions that may be asked to elicit specific information. For example, in the section on reinforcing consequences, this prompt is included: *What do peers say and do when the interfering behavior occurs?* Other prompts are more general. For example, in the section on motivational triggers for the interfering behavior, this prompt is included: *Consider deprivation of sleep, food, or other basic needs; divided/diverted attention; and restricted access to reinforcers.* In this example, interviewers who suspect that specific MOs may be contributing to interfering behaviors should formulate

appropriate questions (e.g., *Do you notice the interfering behavior more often when the student comes to school after a poor night of sleep?* or *How does the student respond when the teacher turns his attention to paperwork or talks with other students in the classroom?*). Given that the variables influencing student behavior tend to be idiosyncratic, interviewers need to be flexible and prepared to frame relevant, individually tailored questions to probe for additional information as needed.

3. *Tables provide a structure for recording interview data; however, it is not necessary to complete every cell in every table.* You will notice that the tables for recording information about MOs, S^Ds, and reinforcing consequences include check-boxes (☐). When interviewees offer information indicative of a particular type of MO, S^D, or reinforcer, you can simply check the relevant box and then input a description. Although you may check multiple boxes (e.g., an interfering behavior may result both in access to social attention and escape from tasks/demands), it is highly unlikely that you would check all boxes. In fact, if you find yourself checking all boxes all the time, then you are probably (a) not isolating the most relevant variables or (b) attempting to gather information about interfering behaviors that are members of different response classes in a single interview.

4. *The BAPS-I is an interview form, not a rating scale.* The form is designed to guide the evaluator in conducting an FBA interview, and it should not simply be given to teachers, parents, or others to fill out on their own.

Remember, the goal of the BAPS-I is to gain a comprehensive understanding of the interfering behavior by “teasing out” the contributing individual and environmental variables. Use it to give your interview structure, but use it flexibly by tailoring questions to the specific referral question and situation!

Interviewing Skills

Conducting an effective FBA interview requires foundational behavior-analytic knowledge, familiarity with specific interview tools, *and* a well-developed repertoire of clinical interviewing skills. It is beyond the scope of this chapter to review the foundations of clinical interviewing, so we encourage you to check out the NASP best practices resource and the classic behavior-analytic study on this topic listed below.

Iwata, B. A., Wong, S. E., Riordan, M. M., Dorsey, M. F., & Lau, M. M. (1982). Assessment and training of clinical interviewing skills: Analog analysis and field replication. *Journal of Applied Behavior Analysis*, 15, 191–203.

Mazza, J. J. (2014). Best practices in clinical interviewing parents, teachers, and students. In P. L. Harrison & T. A. Thomas (Eds.), *Best practices in school psychology: Data-based and collaborative decision making* (pp. 317–330). Washington DC: National Association of School Psychologists.

A BAPS Model Refresher

The BAPS-I is directly aligned with the BAPS model presented in Chapter 6, and a solid grasp of this conceptual model is essential for conducting a valid interview. To *prime* the

interview process, we offer the following condensed version of the essential components of the BAPS model.

- *Context* refers to the physical setting, social circumstances, or types of activities in which interfering behavior occurs. *Examples*: circle time, toileting routines, the cafeteria, independent seatwork.
- *Personal characteristics* include traits, sensitivities, beliefs, and/or values that contribute to interfering behavior. *Examples*: impulsivity, short attention span, sensitive to loud noises, sensitive to peer evaluation, polarized thinking, low frustration tolerance, disorganized, rigid/inflexible.
- *Skill delays/deficits* include weaknesses in communication, academic, executive, adaptive living, social, and/or self-management skills. *Examples*: delays or deficits in reading fluency, functional communication skills, conflict resolution skills, response inhibition, flexible thinking, time management, goal-directed persistence.
- *Motivational triggers (MOs)* include antecedent stimuli and events that (1) momentarily alter the effectiveness of reinforcing consequences and (2) evoke interfering behaviors. *Examples*: deprivation of food or sleep, divided/diverted attention, restricted access to preferred items/activities, presentation of difficult tasks, physical pain, unpleasant thoughts and emotions.
- *Sources of reinforcement (S^D s)* refer to antecedent stimuli (e.g., persons, items, or locations) that signal the availability of reinforcing consequences and occasion interfering behavior. *Examples*: teacher with a history of removing work expectations contingent on interfering behavior, classmates who provide attention for social antics, close proximity of a preferred object.
- *Interfering behavior* refers to the specific responses that interfere with a student's progress. *Examples*: physical aggression (hair pulling, slapping, and biting), vocal opposition (saying no or vocally refusing to complete a task), off-task behavior (manipulating objects unrelated to the task, gazing out the window, etc.).
- *Reinforcing consequences* are the events that follow and strengthen interfering behaviors. Reinforcing consequences may be classified as (1) positive (access) or negative (avoidance/escape) and (2) socially mediated (delivered by others), individually mediated (directly obtained), or automatic (sensory). *Examples*: access to peer attention, access to preferred items, escape from tasks, sensory stimulation, arousal reduction.
- *Parameters of reinforcement* are the dimensions of reinforcement (schedule, quality, magnitude, and timing) that influence its strengthening effects (robustness). *Examples*: accessing highly preferred toys (quality) for 1–5 minutes (magnitude), immediately following aggression toward peers (timing) approximately 25% of the time (schedule).

Behavioral Stream Interview

A less structured type of interview, the BSI, prompts interviewees to describe one or more specific behavioral incidents. It is designed to elicit detailed information about the many

interacting antecedent and consequence variables that influence the occurrence of interfering behavior. Remember, these variables are not stagnant. The ongoing flow of behavior and related stimuli are comparable to a river—sometimes a stream that gently meanders through a meadow, and at other times a raging torrent rushing through the mountainous canyons. The BSI helps to identify the dynamic interactions between these variables by determining the sequence of events that unfold during a behavioral incident. Whereas typical A-B-C descriptions are like photographs that capture a single event at one moment in time, BSI-generated descriptions of behavior are like videos that capture an entire sequence of events unfolding in real time.

Example of a BSI

In this case example, the school psychologist has worked with Robert, a fourth-grade student, for a number of years. The school psychologist is very familiar with Robert's personal characteristics, skill delays/deficits, and school programming and decides to conduct a BSI with the program administrator and a staff member (Jeff) to gather more specific information about the variables influencing Robert's aggressive behaviors. The following interview transcript begins after the school psychologist explains the process of the BSI.

PSYCHOLOGIST: I understand that Robert has been showing a marked increase in aggressive behaviors over the past week. Could you briefly describe those behaviors?

ADMINISTRATOR: Sure . . . he has exhibited several occurrences of severe aggression.

PSYCHOLOGIST: Could you give me an example? What did the aggression look like?

ADMINISTRATOR: Well . . . he pushes other people forcefully and strikes them like this . . .
(Makes a fist and demonstrates the action of striking another person with his fist and forearm.)

PSYCHOLOGIST: OK. So, he pushes and strikes others using his fist and forearm. Have you seen any other forms of aggression?

ADMINISTRATOR: No . . . that's pretty much it.

JEFF: I agree . . . that sums it up.

PSYCHOLOGIST: All right. Then I'd like to review two to three recent examples of Robert's aggressive behavior. Let's start with the most recent incident of aggressive behavior. Tell me what happened.

ADMINISTRATOR: Well, it happened out of the blue. Robert was having a really great day and just before lunch . . . 11:15 A.M. . . . he hit Jeff on the back. The incident resulted in a 48-minute physical restraint.

PSYCHOLOGIST: Let's back up . . . What was going on before the incident of aggression? What was Robert doing and who was around right before the aggressive incident?

JEFF: Robert and I were outside playing pass with a football . . . he really likes to throw the football around. Another staff person, Sandy, came outside and yelled, "It's time for

lunch.” I told Robert that we needed to go inside and wash our hands before lunch . . . then he hit me.

PSYCHOLOGIST: Did anything else happen before he hit you? Think back . . . after you prompted Robert to go inside to wash his hands before lunch, did anything else happen?

JEFF: Yes, now that I think of it . . . yeah . . . I told Robert that we needed to stop playing pass with the football, and he threw the ball over the playground fence. I put my arm on his shoulder and said, “We’ll have to get the ball after lunch. Come on now . . . it’s time for lunch. We need to go in and wash our hands.” That’s right when he hit me.

PSYCHOLOGIST: Would you describe how he hit you?

JEFF: He screamed “AHHHHH,” clenched his fist, and then used a chopping motion to hit me two or three times on my arms and chest. It wasn’t very hard, but it met the criteria for aggressive behavior.

PSYCHOLOGIST: What did you do next?

JEFF: I followed the behavioral support protocol and delivered a verbal reprimand . . . “Robert, no hitting!” That’s when he hit me again . . . three or four more times. The hits were much harder, so I called for support. He kept trying to hit me, but Sandy and Jorge came in less than a minute. We used a protective emergency restraint, and it took 48 minutes for Robert to calm down.

In this example, it is clear that the behavioral incident involved a stream of events with interacting antecedents and consequences. The BSI allowed the psychologist to capture and understand the flow of events. It also exposed the fact that the consequence for one response functioned as the antecedent for the subsequent response, and so on. The descriptions generated during the BSI captured the complexity of an actual incident. And by giving full consideration to the evolution and interaction of variables during several behavioral incidents, the psychologist is much less likely to arrive at faulty hypotheses about the function(s) of Robert’s aggressive behavior.

BSI Recording

To summarize information gathered during a BSI, it is helpful to map out the information using the format shown in Figure 7.2. Essentially, the interviewer and interviewee reconstruct a complete record of each behavioral incident using a series of boxes and arrows to show how actions led to reactions throughout the incident. The process does not require the interviewer to label variables as antecedents or consequences. Instead, the focus is on describing the sequence of events as the behavioral incident unfolds. It really is a matter of recording from a “he said/he did” or “she said/she did” perspective.

The sample behavioral stream record in Figure 7.2 summarizes a behavioral incident involving an elementary student, Seth, who displays a set of behaviors referred to as “tantrum behavior.” For this student, tantrum behavior was defined as a response set including two or more of the following: verbal opposition, swearing, throwing or damaging materials, and physically pushing others. The events captured in the BSI record unfolded during a general education math class when independent math assignments were presented.

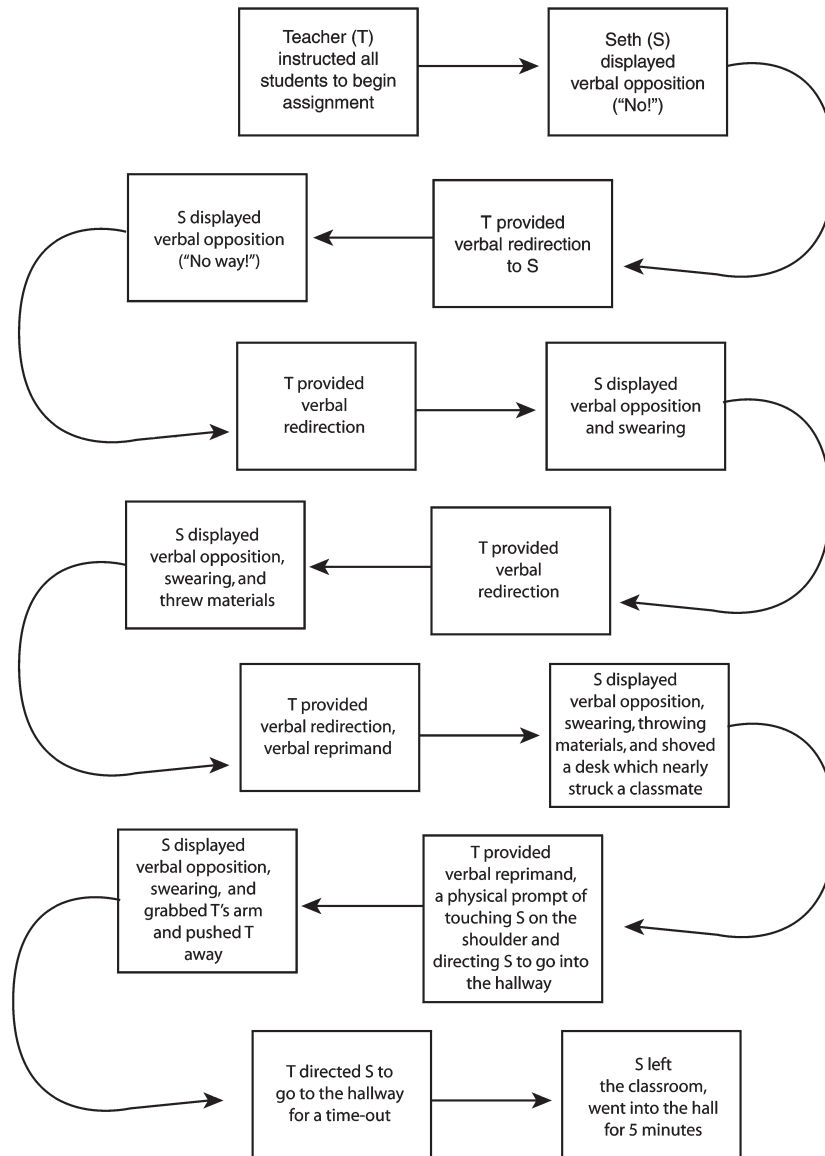
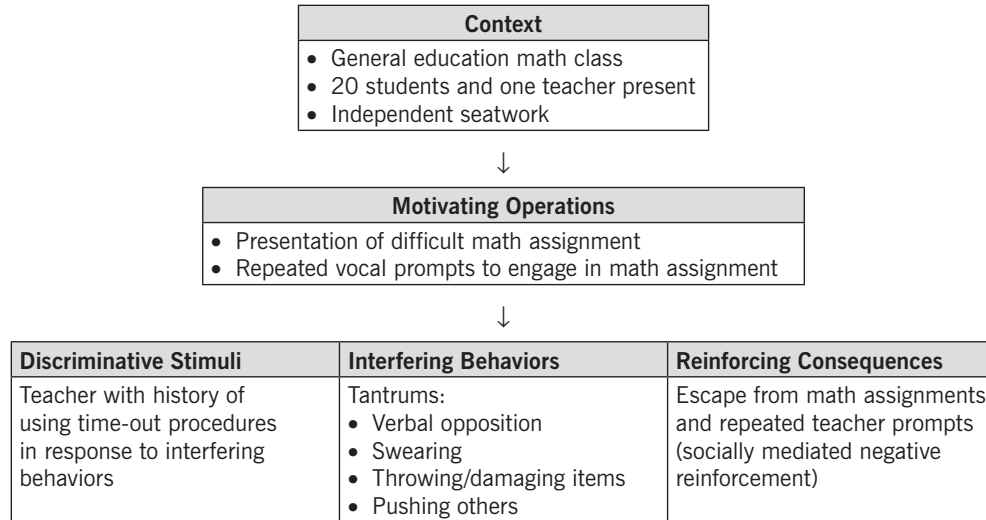


FIGURE 7.2. BSI recording.

BSI Interpretation

After completing a BSI and generating a behavior stream record, evaluators may interpret the data using the IBFAF (Form 8.2 on page 161), which is described in detail in Chapter 8. Figure 7.3 shows the translation of data from the behavior stream record for Seth's tantrum behavior (Figure 7.2) into a conceptual map of hypothesized functional relations between MOs, S^ps, reinforcing consequences, and tantrum behavior. The BSI clearly illustrated the topography and hierarchical sequence of Seth's interfering behaviors, which became increasingly intense over the course of the incident. The BSI also offered a clear description of the consequences for Seth's tantrum behavior. By engaging in these behaviors, Seth avoided completing the assignment and ultimately escaped from the entire aversive situa-

**FIGURE 7.3.** IBFAF results for Seth.

tion (i.e., the assignment and the teacher prompts). Finally, the functional antecedents (MOs and S^Ds) were surmised on the basis of information about the events that preceded and followed the tantrum behaviors.

CAUTION!!!

Conducting a BSI facilitates the development of a hypothesis about the antecedents and reinforcing consequences associated with *one* occurrence of interfering behavior. *However, it is important not to rely on data from a single behavioral incident to draw conclusions about the function(s) of an interfering behavior.* Serious mistakes can occur when hypotheses are generated on the basis of inadequate data.

For example, after using the BSI to gather information about a recent tantrum exhibited by Seth, the evaluator hypothesized that repeated verbal cues from the teacher may evoke tantrums. However, after directly observing and recording data during multiple tantrum episodes, the evaluator discovered that other types of repeated verbal cues (e.g., to get out materials, line up for lunch, and eat a snack) did not evoke tantrums. Instead, tantrums occurred most consistently when repeated verbal cues were delivered along with math assignments. Without conducting descriptive FBA procedures, the evaluator simply could not determine whether repeated verbal cues, math assignments, neither variable, or both variables served as functional MOs for tantrum behavior.

SUMMARY

Indirect FBA is an efficient, but often insufficient, method for assessing the function(s) of interfering behavior. Reviewing school records, administering rating scales, and conducting interviews provides important information about antecedent, individual, and consequence variables associated with interfering behavior. A strength is that these methods enable evaluators to gather information about a wide range of variables, both current and historical, that may contribute to interfering behaviors; the scope of assessment is not limited exclu-

sively to what can be observed in the moment. The weakness is that indirect FBA relies on the reports and recollections of others, which are notoriously unreliable.

The indirect FBA procedures described in this chapter provide a flexible means to gather and summarize comprehensive information about students and the variables influencing their behavior. In some situations, indirect FBA procedures are in and of themselves sufficient for assessing and understanding behavioral function. For example, in intensive applied behavior-analytic educational settings, the availability of highly trained teachers to interview and reliable existing data from routine data collection and observation may render an indirect FBA sufficient for the purposes of reevaluation. In other situations, an indirect FBA may be the only immediate, feasible option. For example, a school psychologist who is requested to conduct an FBA of aggressive behaviors *before* a suspended student is permitted to return to school may not have an opportunity to conduct direct observations. In these cases, practitioners may decide to conduct a comprehensive indirect FBA, as described in this chapter, and use results that are clear and consistent to design function-based interventions. If the intervention is implemented and proves to be successful, then the assessment process may be considered validated. If, however, indirect FBA results are ambiguous and inconsistent or interventions are ineffective, then additional descriptive assessments and/or experimental analyses are essential. The next two chapters focus on these additional procedures.

Behavior-Analytic Problem-Solving Interview (BAPS-I)

Student's Name: _____ DOB: _____

School: _____ Date: _____

Evaluator: _____

The purpose of this semistructured interview is to identify and describe (1) student-specific interfering behavior(s) and (2) the individual and environmental variables that contribute to, evoke, occasion, and maintain interfering behavior(s). The interview is based on the eight-term behavior-analytic problem-solving model (Chapter 6). Each section of this form includes questions and prompts to guide the interviewer in gathering relevant information from caregivers, teachers, or others familiar with the student.

IDENTIFY AND DESCRIBE PERSONAL STRENGTHS

1. What are [the student's] most significant strengths?

- a. *Prompt: Identify communication, academic, executive, adaptive living, social, and/or self-management skills in the student's current repertoire.*

Personal Strength	Description

(continued)

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Behavior-Analytic Problem-Solving Interview (BAPS-I) (page 2 of 6)

IDENTIFY AND DESCRIBE THE INTERFERING BEHAVIOR

Note: After identifying the interfering behavior, focus all interview questions on this target. Conduct separate interviews, using separate forms, for each target response class.

1. **What behavior interferes most with [the student's] day-to-day functioning? What behavior is most concerning or problematic?**
 - a. *Prompt: Identify the behavior(s) that led to the referral.*
 - b. *Prompt: If multiple behaviors, prioritize one target at a time for assessment.*
2. **Describe the interfering behavior. What does it look like?**
 - a. *Prompt: Ask for specific examples or demonstrations.*
3. **How does an “episode” of the interfering behavior typically unfold? Are there early signs or predictable sequences of behavior during an episode?**
 - a. *Prompt: If applicable, identify the response class hierarchy.*

Interfering Behavior	Definition and Examples

Response Class Hierarchy	Examples

IDENTIFY AND DESCRIBE THE CONTEXT

Note: Focus on the context relevant to occurrences of the target interfering behavior.

1. **When is the interfering behavior most likely to occur? Are there particular locations, settings, or activities when the interfering behavior is most probable?**
 - a. *Prompt: Identify specific physical locations, social circumstances, or types of activities that are challenging for the student.*
2. **What behaviors are expected of students in that context?**

Description of Context	Description of Expected Behaviors

(continued)

Behavior-Analytic Problem-Solving Interview (BAPS-I) (page 3 of 6)

IDENTIFY AND DESCRIBE MOTIVATIONAL “TRIGGERS” (MOs)

Note: Focus on the MOs relevant to occurrences of the target interfering behavior.

1. **What seems to trigger the interfering behavior? What could I do right now to make the interfering behavior happen?**
 - a. *Prompt: Identify relevant conditions of deprivation or restricted access. Consider deprivation of sleep, food, or other basic needs; divided/diverted attention; and restricted access to reinforcers.*
 - b. *Prompt: Identify aversive stimuli/events that evoke interfering behavior. Consider physical pain and discomfort, unpleasant social interactions, nonpreferred task demands, and unpleasant thoughts and emotions.*

Potential MO	Description
☐ Deprivation of physical wants/needs	
☐ Divided or diverted attention	
☐ Restricted/denied access to reinforcers	
☐ Physical pain/discomfort	
☐ Unpleasant thoughts/emotions	
☐ Nonpreferred tasks/activities	
☐ Aversive social situations	

IDENTIFY AND DESCRIBE SOURCES OF REINFORCEMENT (S^Ds)

Note: Focus on the S^Ds relevant to occurrences of the target interfering behavior.

1. **Who is typically present when the interfering behavior occurs? Is the interfering behavior more or less likely to occur when certain people are present?**
 - a. *Prompt: Does the interfering behavior occur more often in the presence of specific peers, teachers, or other school staff? If so, whom?*
2. **Are there specific items or places that seem to “trigger” the interfering behavior?**
 - a. *Prompt: Does the interfering behavior occur reliably when certain objects are present or in specific physical locations/settings?*

Potential S ^D	Description
☐ Persons	
☐ Items/Locations	

(continued)

Behavior-Analytic Problem-Solving Interview (BAPS-I) (page 4 of 6)

IDENTIFY AND DESCRIBE REINFORCING CONSEQUENCES AND PARAMETERS OF REINFORCEMENT

Note: Focus on the consequences relevant to occurrences of the target interfering behavior.

Assess for Socially Mediated Reinforcement

1. **How do you and others typically respond to the interfering behavior?**
2. **What seems to work to distract or deescalate [the student] when the interfering behavior occurs?**
 - a. *Prompt: Consider access to peer and adult attention (visual, vocal, and physical). What do adults say and do when the interfering behavior occurs? What do peers say and do when the interfering behavior occurs?*
 - b. *Prompt: Consider access to edibles, tangibles, and activities. Does the student ever get access to things or activities he/she wants by engaging in the interfering behavior?*
 - c. *Prompt: Consider avoidance or escape from aversive tasks/activities/situations. Does the interfering behavior allow the student to get out of task expectations? Is the interfering behavior followed by changes in task expectations (e.g., modifications or alternative tasks)? Does the interfering behavior lead to the student's removal from the setting? Do people "back off" when the interfering behavior occurs?*

Potential Socially Mediated Reinforcing Consequence	Description
☐ Access to social attention	
☐ Access to edibles, tangibles, activities	
☐ Escape from tasks/activities	
☐ Escape from other aversive situations	

Assess for Individually Mediated Reinforcement

1. **Does [the student] ever gain access to desired items, activities, or locations by engaging in the interfering behavior?**
 - a. *Prompt: Determine if the interfering behavior results in direct access to preferred edibles, tangibles, activities, or locations without help or mediation from others.*
2. **Does [the student] ever escape from items, activities, or locations he/she does not like by engaging in the interfering behavior?**
 - a. *Prompt: Determine if the interfering behavior results in direct escape from aversive edibles, tangibles, activities, or locations without help or mediation from others.*

Potential Individually Mediated Reinforcing Consequence	Description
☐ Access to edibles, tangibles, activities, or locations	
☐ Escape from activities, locations, or other aversive stimuli	

(continued)

Behavior-Analytic Problem-Solving Interview (BAPS-I) (page 5 of 6)

Assess for Automatic Reinforcement

1. Does the interfering behavior ever occur when [the student] is alone?
2. Does the interfering behavior appear to result in sensory consequences that [the student] enjoys?
 - a. Prompt: Assess for “self-stimulating” functions: visual, auditory, olfactory, or tactile. Does the interfering behavior occur more frequently when [the student] is “bored” or not directly engaged with people and preferred items/activities?
3. Does the interfering behavior seem to provide a “release” or sense of relief for [the student]?
 - a. Prompt: Assess for arousal reduction. Does the interfering behavior typically occur when [the student] is tense, anxious, frustrated, or angry? Does the interfering behavior occur when [the student] is in pain?

Potential Automatic Reinforcing Consequence	Description
☐ Sensory stimulation	
☐ Arousal reduction	

Assess for Parameters of Reinforcement

Note: List each identified reinforcing consequence in the table below and describe the associated parameters of reinforcement to estimate the strength/robustness of consequences relevant to the interfering behavior.

1. Tell me more about situations when the interfering behavior results in [reinforcing consequence]. What exactly does that look like?
 - a. Prompt: Estimate schedule. How often does the interfering behavior “work”?
 - b. Prompt: Estimate quality. How valuable is the reinforcing consequence for [the student]?
 - c. Prompt: Estimate magnitude. How long does [the student] get the reinforcing consequence? How much of the reinforcing consequence does [the student] get?
 - d. Prompt: Estimate timing. Is the consequence immediate or delayed?

Possible Reinforcing Consequence	Parameters: Schedule, Quality, Magnitude, Timing

IDENTIFY AND DESCRIBE PERSONAL CHARACTERISTICS

Note: Focus on personal characteristics relevant to occurrences of the target interfering behavior.

1. What personal characteristics make it difficult for [the student] to be successful in [the specific context(s) associated with interfering behaviors]?
 - a. Prompt: Identify relevant patterns of feeling or thinking.
 - b. Prompt: Identify relevant biological/medical conditions.

(continued)

- c. *Prompt: Identify relevant diagnoses.*
- d. *Prompt: Identify relevant cultural factors.*

Traits	
Sensitivities	
Beliefs	
Values	
Other	

Note: Focus on the delays/deficits relevant to occurrences of the target interfering behavior.

1. **What skills does [the student] need to learn to be more successful in [specific context associated with interfering behavior]?**
 - a. *Prompt: Consider communication, academic, executive, adaptive living, social, and self-management skills.*
2. **Are there particular skill deficits that contribute to the interfering behavior?**
 - a. *Prompt: If the identified deficit was a strength, would the interfering behavior be less likely to occur?*

Communication Skills	
Academic Skills	
Executive Skills	
Adaptive Living Skills	
Social Skills	
Self-Management Skills	

[illegible]